UNNS-U2-RN

Ranked-Neighbour Dynamics and Graph-Local Observability
in the Reversible SELECT-II Graph

UNNS Research Collective

Version v001 (separate addendum; UNNS-U2-TI v001 and UNNS-U2-TI-DYN v001 and all earlier releases are unchanged)

Abstract

We ask what the arithmetic graph G (edges $\{kN, k(N+1)^2\}$) exposes through its own local structure, with no sequence-specific controller. **RN-0** fixes a certification policy: a neighbourhood or degree is *certified* only if every prime factor used is proved prime; runs lacking this are flagged. **RN-1** proves closed forms for seven ranked-neighbour operators and determines the full dynamics of six of them: the largest neighbour gives $x \mapsto (x+1)^2$; the second-largest is conjugate to it after one step; inverse-side and smallest-neighbour rules descend to squarefree numbers in at most $\log_4 x$ steps, and the smallest-neighbour rule then enters the 2-cycle $\{s, 4s\}$. Forward moves never land on a prime, and a prime's only neighbours are $4p$ and $(p+1)^2$; hence every one of these rules visits at most one distinct prime. **RN-2** classifies 840 orbits of a registered 60-rule grammar: 93 cyclic, 405 escaping, 342 stuck, none uncertified. Rules reading vertex degree reach up to 6 distinct primes in finite orbits, with no theorem behind it. **RN-3** gives an observability ledger. Recognizing a prime vertex locally is a theorem; generating infinitely many primes by a graph-local rule is not shown, and the proved results are negative for the rank-based family.

1 Scope and evidence levels

Every statement carries one label. **Proved** (elementary argument in this note, no novelty claim). **Verified-bounded** (exact-integer check over a stated range; corroboration, not proof). **Registered observation** (outcome of the pre-registered run). **Exploratory** (looked at after results were seen, or not registered). **Prior art**. Registered code and configuration are SHA-256 pinned; the verifier recomputes all 840 orbits and compares them with the stored results.

2 RN-0: certification

Policy. Factorization is exact integer arithmetic (trial division, Pollard ρ, Miller–Rabin). A composite witness is always a proof. A *primality* claim is a proof only for $n < 3.317 \times 10^{24}$ (the 13-base deterministic range, Sorenson–Webster); above that it is flagged *uncertified*. A neighbourhood (and hence a degree or a rank) is *certified* iff every prime factor used was certified. Prime *certification* is not complete *factorization*: a number can be fully factored with only probable-prime cofactors. Both are tracked; the run flag is the conjunction.

Checks (verified-bounded). Primality agrees with a sieve for all $0 \le n \le 2{,}000{,}000$ and rejects four strong pseudoprimes. 400 random factorizations up to 10^{24} equal SymPy's. Neighbour sets equal a definition-level engine (no factoring) for $x \le 20{,}000$, and equal the brute-force edge set of the grammar (all edges with ends $\le 10^6$) for $x \le 999$; degrees match. The Mersenne prime

$2^{89}-1$ (above the bound) is correctly flagged uncertified, and a run started there is flagged. A scan finds no float arithmetic except one exponent `.5` in the one-line bound of the sieve for primes below 2000 (the resulting list is re-verified exactly). All 840 registered orbits stay below 10^{18}, so 0 are uncertified; the flag is a safeguard for larger runs.

3 RN-1: structural theorems

Throughout, for $x \geq 1$: forward neighbours $F(x) = \{x(d+1)^2/d : d \mid x\}$ and inverse neighbours $I(x) = \{x(q-1)/q^2 : q \geq 2,\ q^2 \mid x\}$. This is Proposition 2.1 of TI-DYN v001, re-derived here.

Proposition 1 (Neighbourhood)**.** *Every edge at x is one of these; $d \mapsto d+2+1/d$ is strictly increasing so the forward outputs are distinct and increase with d; $q \mapsto (q-1)/q^2$ is strictly decreasing on $q \geq 2$ so inverse outputs are distinct and decrease with q; $F(x) > x > I(x)$ elementwise.*

Proof. $x = kN$ gives $d = N$; $x = k(N+1)^2$ gives $q = N+1$. Differences: $f(d{+}1) - f(d) = 1 - \frac{1}{d(d+1)} > 0$, and $g(q) - g(q{+}1) = \frac{q^2-q-1}{q^2(q+1)^2} > 0$ for $q \geq 2$. $f(d) > 1 > g(q)$. □

Write $x = a^2 b$ with b squarefree (a unique), $p = \mathrm{spf}(x)$ and $p_2 =$ the least prime with $p_2^2 \mid x$.

Theorem 2 (Ranked closed forms; Proved)**.**

operator	*divisor/square index*	*value*	*defined*
largest forward	$d = x$	$(x+1)^2$	*always*
smallest forward	$d = 1$	$4x$	*always*
2nd-largest forward	$d = x/p$	$(x+p)^2/p$	$x > 1$
2nd-smallest forward	$d = p$	$x(p+1)^2/p$	$x > 1$
largest inverse	$q = p_2$	$x(p_2-1)/p_2^2$	x *not squarefree*
smallest inverse	$q = a$	$b(a-1) =: \sigma(x)$	x *not squarefree*
smallest overall	—	$\sigma(x)$ *if* $a > 1$, *else* $4x$	*always*

Largest overall equals largest forward, and second-largest overall equals second-largest forward for $x > 1$ (all forward neighbours exceed all inverse ones).

Proof. Forward rank follows the divisor order (Prop. 1); the largest proper divisor of $x > 1$ is x/p and the second-smallest divisor is p. $q^2 \mid x$ iff $q \mid a$; the smallest $q \geq 2$ is a prime (if $q^2 \mid x$ and $\ell \mid q$ prime then $\ell^2 \mid x$), namely p_2, and the largest is a; $x(a-1)/a^2 = b(a-1)$. □

Verified-bounded: each row equals the definition-level rank for all $x \leq 20000$ (0 mismatches).

Lemma 3 (Forward landings are composite; prime neighbourhoods; Proved)**.** *(i) Every forward neighbour $y = (x/d)(d{+}1)^2$ is composite. (ii) The neighbours of a prime p are exactly $4p$ and $(p{+}1)^2$. (iii) A vertex x has a prime neighbour iff $x = 4p$ or $x = (p+1)^2$ for a prime p; that neighbour is p (the only one). (iv)* $\deg(1) = 1$*, primes have degree 2, no vertex has degree 3, composites have degree ≥ 4, and* $\deg x = \tau(x) + \#\{q \geq 2 : q^2 \mid x\}$.

Proof. (i) $(d+1)^2 \mid y$ with $d+1 \geq 2$, and $y \geq 4$. (ii) $F(p) = \{4p, (p+1)^2\}$, $I(p) = \emptyset$. (iii) By (i) a prime neighbour is inverse: $y = kN$ with $x = k(N+1)^2$. Prime needs $k = 1, N$ prime (so $x = (N+1)^2$, $y = N$) or $N = 1, k$ prime ($x = 4k$). Conversely these are neighbours. (iv) Prop. 1; a composite has $\tau \geq 3$, and $\tau = 3$ means $x = \ell^2$ with the extra $q = \ell$. □

Verified-bounded: (i)–(iii) for $x \leq 20000$, (iv) for $x \leq 10^6$. The prime–degree equivalence is also TI-DYN v001 Theorem 4.1 (re-proved).

Corollary 4 (Local visibility of primes; Proved)**.** *For $x \geq 1$, $\min_{y \sim x} \deg y = 1$ iff $x = 4$, $= 2$ iff $x \in \{4p, (p+1)^2\}$ $(x \neq 4)$, and is ≥ 4 otherwise. A degree-3 target is never available. So a degree-reading selector sees a prime neighbour only from two vertex families.*

Proof. Degree 1 occurs only at 1, a neighbour only of 4. Then Lemma 3. □

Theorem 5 (Dynamics of the stationary rank rules; Proved)**.** *Let $x_{j+1} = \Sigma(x_j)$.*

1. ***Largest neighbour.*** *$x_{j+1} = (x_j + 1)^2$. Strictly increasing, non-periodic, $x_j \geq 4^{2^{j-1}}$ $(j \geq 1)$.*
2. ***Smallest forward.*** *$x_j = 4^j x_0$.*
3. ***Second-largest.*** *$x_0 = 1$ is stuck. For $x_0 > 1$ all x_j $(j \geq 1)$ are even and $u_j = x_j/2$ satisfies $u_{j+1} = (u_j + 1)^2$: the orbit is conjugate to (1) after at most one step.*
4. ***Inverse largest/smallest.*** *Every step has $y \leq x/4$; the orbit stops (undefined rank,* stuck*) at a squarefree vertex after at most $\lfloor \log_4 x_0 \rfloor$ steps.*
5. ***Smallest overall.*** *The orbit follows σ to a squarefree s (at most $\lfloor \log_4 x_0 \rfloor$ steps) and is then periodic: $s \to 4s \to s \to \cdots$.*

Proof. (1) Theorem 2; $(x+1)^2 \geq x^2$ and $x_1 \geq 4$. (2) $d = 1$. (3) For even x, $(x+2)^2/2 = 2(x/2+1)^2$. For odd $x > 1$, x/p is odd, so $y = p(x/p+1)^2 = 4p\,((x/p+1)/2)^2$ is even. (4) $(p-1)/p^2 \leq \frac{1}{4}$ and $(a-1)/a^2 \leq \frac{1}{4}$; the rule is undefined iff there is no inverse neighbour iff x is squarefree. (5) On non-squarefree x it is σ. For squarefree s the minimum is $4s$ (there are no inverse neighbours), and since s is squarefree the decomposition $4s = a^2 b$ has $a = 2$, $b = s$ (for odd s clearly; for $s = 2t$ with t odd, $4s = 2^2 \cdot s$ and s is squarefree), so $\sigma(4s) = s \cdot (2-1) = s$. □

Verified-bounded: (1)–(5) against the engine, 13626 orbit comparisons (starts 1–1500 and the 14 registered starts, 0 mismatches), and 112 no-backtrack variant comparisons; descent and the 2-cycle for all $x_0 \leq 100{,}000$ (longest descent 5 steps).

Corollary 6 (At most one prime; Proved)**.** *Along any orbit of the rules in Theorem 5 and of the 2nd-smallest-forward rule, at most one distinct prime is visited.*

Proof. Forward moves land on composites (Lemma 3(i)). A prime has no inverse neighbour, so inverse-only orbits can reach a prime only as the final vertex; their earlier vertices are non-squarefree, hence composite. In (5) the transient vertices are non-squarefree and the cycle $\{s, 4s\}$ contains at most one prime. □

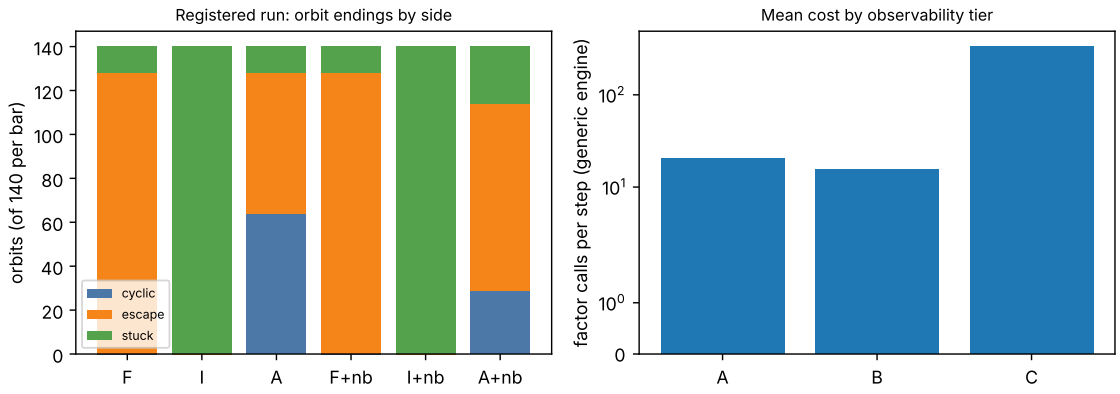
Remark 7 (Negative results, Proved). **N1.** A forward-side rule never visits a prime after its first vertex, and is strictly increasing, so it never cycles. **N2.** An inverse-side rule is strictly decreasing, so every orbit is finite and ends stuck. **N3.** A rank-3 forward rule is stuck exactly at $x_0 \in \{1\} \cup$ primes, and nowhere later. **N4.** If a rule enters a prime p at time $t+1 \geq 1$, then $x_t \in \{4p, (p+1)^2\}$ and the move was inverse (Lemma 3); to visit infinitely many distinct primes a graph-local rule must make infinitely many inverse moves and pass infinitely often through such vertices. No rank-based rule here does. **N5.** Smallest overall is not a prime generator: its attractor $\{s, 4s\}$ is fixed by the start.

Open within this note. The dynamics of 2nd-smallest forward ($x \mapsto x(p+1)^2/p$) beyond one step is observed (Section 4) but not proved. The terminal squarefree numbers of the descent maps are a number-theoretic statistic, not a theorem (exploratory fractions below).

4 RN-2: the registered grammar

Grammar GL-2 (registered). Side $\in \{F, I, A\}$ (forward, inverse, all); either *rank* (top/bottom, $k \leq 3$) or *degree* (min/max degree of the candidate, tie to the larger/smaller value); optionally *no-backtrack* (previous vertex removed from the eligible set before ranking). 60 rules, 14 starts $\{1, \ldots, 10, 12, 16, 24, 36\}$, horizon 400, escape bound 10^{18}. **Undefined rank is stuck; no substitute rule is ever used.** No success criterion is defined: this is a classification study. Outcome counts (cyclic/escape/stuck) per selector, with and without no-backtrack, and the largest number of distinct primes in any of its orbits:

selector	no-backtrack off	on	max distinct primes
F top-1	0/14/0	0/14/0	1
F top-2	0/13/1	0/13/1	1
F top-3	0/9/5	0/9/5	1
F bottom-1	0/14/0	0/14/0	1
F bottom-2	0/13/1	0/13/1	1
F bottom-3	0/9/5	0/9/5	1
F degmin/hi	0/14/0	0/14/0	1
F degmin/lo	0/14/0	0/14/0	1
F degmax/hi	0/14/0	0/14/0	1
F degmax/lo	0/14/0	0/14/0	1
I top-1	0/0/14	0/0/14	1
I top-2	0/0/14	0/0/14	1
I top-3	0/0/14	0/0/14	1
I bottom-1	0/0/14	0/0/14	1
I bottom-2	0/0/14	0/0/14	1
I bottom-3	0/0/14	0/0/14	1
I degmin/hi	0/0/14	0/0/14	1
I degmin/lo	0/0/14	0/0/14	1
I degmax/hi	0/0/14	0/0/14	1
I degmax/lo	0/0/14	0/0/14	1
A top-1	0/14/0	0/14/0	1
A top-2	0/13/1	0/13/1	1
A top-3	0/9/5	0/9/5	1
A bottom-1	14/0/0	5/0/9	2
A bottom-2	13/0/1	13/0/1	1
A bottom-3	9/0/5	9/0/5	1
A degmin/hi	14/0/0	0/13/1	2
A degmin/lo	14/0/0	2/8/4	6
A degmax/hi	0/14/0	0/14/0	1
A degmax/lo	0/14/0	0/14/0	1



Registered observations. (a) Every orbit of the rules covered by Theorem 5 equals its closed form (Verified-bounded above). (b) All inverse-side rules end stuck and all forward-side rules escape or are stuck at the start: N1–N3. (c) Smallest-overall without no-backtrack: all 14 orbits are 2-cycles $\{s, 4s\}$ (Thm. 5(5)). With no-backtrack it is not stationary: it gave cycles of period 21 (e.g. $5, 20, 80, 15, 60, 240, 45, 10, 40, 160, 30, 120, 480, 90, 20, 5, 36, 8, 2, 9, \dots$) and stuck orbits; this is observed, not proved. (d) Degree-min over all neighbours, without no-backtrack, gave 2-cycles for all 28 orbits; with no-backtrack and ties to the smaller value it gave 2 cyclic, 8 escaping and 4 stuck orbits. From $x_0 = 2$ it visits $2, 9, 36, 5, 20, 125, 500, 45, 10, 121, \dots$ and reaches the primes $5, 31, 6763, 30501131$ before leaving the bound; from $x_0 = 6$ it visits six distinct primes $2, 5, 13, 31, 6763, 30501131$. These primes are not the next-prime sequence and no law is claimed. All orbits stay below 10^{18} and are certified.

Prior art. The map $x \mapsto (x+1)^2$ is OEIS A004019 (Sloane; asymptotics $\lfloor c^{2^n} \rceil$ via Aho–Sloane). One web search found no source defining this graph; this is not a prior-art review, and no priority is claimed.

Exploratory. Following σ to the squarefree attractor from $x_0 \le 200{,}000$, the attractor is prime for a fraction 0.136 of starts (largest-inverse descent: 0.154). No interpretation is offered.

5 RN-3: observability and cost ledger

Observation channels: **A** only the vertex x (closed form without factoring: largest and smallest forward, i.e. top-1 on F/A, bottom-1 on F); **B** the factorization of x (smallest prime factor or square part: 2nd-largest/smallest forward, top-1 and bottom-1 inverse, bottom-1 on A); **C** the complete neighbour set, and for degree rules the factorization of every candidate neighbour (composed from the pieces, not refactored). Counts are measured in *this* Python implementation with caches cleared per orbit, three starts $\times 60$ steps; "cf" is the closed-form evaluator of Theorem 2 where one exists.

selector (no-backtrack off)	tier	factor calls/step	ρ iterations/step	cf calls/step
F top-1	A	24.2	390.8	0.0
F top-2	B	22.9	240.6	1.0
F top-3	C	43.5	724.7	–
F bottom-1	A	13.5	778.4	0.0
F bottom-2	B	16.1	1002.5	1.0
F bottom-3	C	39.5	1628.0	–
F degmin/hi	C	9.0	428.9	–
F degmin/lo	C	8.7	1315.7	–
F degmax/hi	C	2197.8	44230.6	–
F degmax/lo	C	1125.0	13233.8	–
I top-1	B	12.0	0.0	1.6
I top-2	C	32.0	0.0	–
I top-3	C	30.5	0.0	–
I bottom-1	B	12.4	0.0	1.6
I bottom-2	C	32.0	0.0	–
I bottom-3	C	29.5	0.0	–
I degmin/hi	C	12.4	0.0	–
I degmin/lo	C	12.4	0.0	–
I degmax/hi	C	13.0	0.0	–
I degmax/lo	C	13.0	0.0	–
A top-1	A	24.2	390.8	0.0
A top-2	B	22.9	240.6	1.0
A top-3	C	43.5	724.7	–
A bottom-1	B	7.2	0.0	1.0
A bottom-2	C	10.1	0.0	–
A bottom-3	C	10.1	0.0	–
A degmin/hi	C	8.4	0.0	–
A degmin/lo	C	7.2	0.0	–
A degmax/hi	C	2197.8	44230.6	–
A degmax/lo	C	1125.0	13233.8	–

Reading: a rule can be tier A as mathematics while the generic neighbour-enumerating engine still pays tier C prices (the table shows 24 factor calls per step for the largest-neighbour rule against 0 for its closed form). Degree-max rules are the most expensive, since each candidate must be factored. These are operation counts, not intrinsic complexity.

6 Findings and limits

Proved	Thm. 2, Lemma 3, Cor. 4, Thm. 5, Cor. 6, N1–N5. All elementary; (largest neighbour) and the degree–prime equivalence are known from TI-DYN v001.
Verified-bounded	the ranges in RN-0/RN-1; 13626 proved-law orbit comparisons.
Registered obs.	Table of 840 orbits; items (a)–(d).
Exploratory	terminal-prime fractions; the prime-rich degree-min orbits.
Open	dynamics of 2nd-smallest forward; periods under no-backtrack; whether any graph-local rule visits infinitely many distinct primes.
Prior art	A004019 only; no priority claimed.

Amendments / disclosures. None to the registration. Two notes: `rn_closed.orbit` (registered, unused) can miss a cycle shorter than four terms, so the verifier uses its own orbit loop with the same closed-form maps; the float exponent in the small-prime sieve is disclosed in RN-0. **Unverified:** behaviour beyond horizon 400 or 10^{18}; rank $k \geq 3$ dynamics; any rule outside GL-2; whether richer observation (e.g. neighbours of neighbours) changes the prime picture.

Consequence. The graph exposes primality exactly (degree 2) but only in the two vertex families adjacent to a prime. Graph-local ranked selectors reproduce elementary dynamics (doubly exponential growth, geometric growth, finite descent, a 2-cycle) and cannot, by Corollary 6, generate

primes. Degree-reading selectors are the only observed route to several primes per orbit; whether that scales is open. Recognizing a prime vertex is a theorem; generating the primes is not shown.